The eigenvalues of $\mathcal{H}$ are 0.38 and 2.62. None of the eigenvalues has a magnitude of one so we are able to continue with the procedure. The eigenvector of $\mathcal{H}$ that corresponds to the eigenvalue outside the unit circle is $\begin{bmatrix} 0.5257 & 0.8507 \end{bmatrix}^T$. We form the corresponding eigenvector matrix as

$$\begin{bmatrix} \Psi_{12} \\ \Psi_{22} \end{bmatrix} = \begin{bmatrix} 0.5257 \\ 0.8507 \end{bmatrix} \tag{7.104}$$

Note that Ψ_{12} is invertible so we are able to continue with the problem. The steady-state Riccati equation solution is

$$\begin{aligned} P &= \Psi_{22}\Psi_{12}^{-1} \\ &= \frac{0.8507}{0.5257} \\ &= 1.62 \end{aligned} \tag{7.105}$$

The steady-state Kalman gain is therefore computed from Equation (7.14) as

$$\begin{aligned} K &= PH^T(HPH^T + R)^{-1} \\ &= \frac{(1.62)(1)}{(1)(1.62)(1) + 1} \\ &= 0.62 \end{aligned} \tag{7.106}$$

which is in agreement with Equation (7.37).

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7.4 KALMAN FILTERING WITH FADING MEMORY

In Section 5.5, we discussed the problem of filter divergence due to mismodeling. That is, if our system model does not match reality, then the Kalman filter estimate may diverge from the true state. Example 5.3 showed how the addition of fictitious process noise can compensate for mismodeling. In this section, we show how to accomplish the same thing with the fading-memory filter. Recall our linear discrete-time system model:

$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \\ E[w_k w_j^T] &= Q_k \delta_{k-j} \\ E[v_k v_j^T] &= R_k \delta_{k-j} \\ E[w_k v_j^T] &= 0 \end{aligned} \tag{7.107}$$

The Kalman filter finds the sequence of estimates $\{\hat{x}_1^-, \cdots, \hat{x}_N^-\}$ that minimizes $E(J_N)$, where J_N is given as

$$J_N = \sum_{k=1}^{N} \left[(y_k - H_k\hat{x}_k^-)^T R_k^{-1}(y_k - H_k\hat{x}_k^-) + \hat{w}_k^T Q_k^{-1}\hat{w}_k\right] \tag{7.108}$$

Note that $\hat{x}$ determines $\hat{w}$ through the system equation, and vice versa. This expression for J_N shows how we could give greater emphasis to more recent data. Instead of finding the filter that minimizes $E(J_N)$, we can find the filter that minimizes $E(\tilde{J}_N)$, where $\tilde{J}_N$ is given as

$$\tilde{J}_N = \sum_{k=1}^{N} \left[(y_k - H_k\hat{x}_k^-)^T \alpha^{2k} R_k^{-1} (y_k - H_k\hat{x}_k^-) + \hat{w}_k^T \alpha^{2k+2} Q_k^{-1} \hat{w}_k\right] \tag{7.109}$$

where $\alpha \geq 1$. The α term in the first part of the cost function means that we are more interested in minimizing the weighted covariance of the residual at recent times (large values of k) than at times in the distant past (small values of k). This will force the filter to converge to state estimates that discount old measurements and give greater emphasis to more recent measurements. The α term in the second part of the cost function is added for mathematical tractability, as we will see in the subsequent development. The second part of the cost function is constant as far as our minimization problem is concerned.

The solution to the minimization of $E(\tilde{J}_N)$ is equivalent to the minimization of $E(J_N)$ (which is the Kalman filter), except that R_k is replaced with $\alpha^{-2k}R_k$ and Q_k is replaced with $\alpha^{-2k-2}Q_k$. The modified Kalman gain can therefore be written as

$$\begin{aligned} K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + \alpha^{-2k} R_k)^{-1} \\ &= \alpha^{2k} P_k^- H_k^T (H_k \alpha^{2k} P_k^- H_k^T + R_k)^{-1} \end{aligned} \tag{7.110}$$

The time update for the estimation-error covariance can be written as

$$\begin{aligned} P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + \alpha^{-2k+2} Q_{k-1}/\alpha^2 \\ \alpha^{2k} P_k^- &= F_{k-1} \alpha^{2k} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \\ &= \alpha^2 F_{k-1} \alpha^{2(k-1)} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \end{aligned} \tag{7.111}$$

The measurement update for the estimation-error covariance can be written as

$$\begin{aligned} P_k^+ &= P_k^- - K_k H_k P_k^- \\ \alpha^{2k} P_k^+ &= \alpha^{2k} P_k^- - K_k H_k \alpha^{2k} P_k^- \end{aligned} \tag{7.112}$$

Now we define $\tilde{P}_k^+$ and $\tilde{P}_k^-$ as

$$\begin{aligned} \tilde{P}_k^+ &= \alpha^{2k} P_k^+ \\ \tilde{P}_k^- &= \alpha^{2k} P_k^- \end{aligned} \tag{7.113}$$

We can then write Equations (7.110), (7.111), and (7.112) as

$$\begin{aligned} K_k &= \tilde{P}_k^- H_k^T (H_k \tilde{P}_k^- H_k^T + R_k)^{-1} \\ \tilde{P}_k^- &= \alpha^2 F_{k-1} \tilde{P}_{k-1}^+ F_{k-1}^T + Q_{k-1} \\ \tilde{P}_k^+ &= \tilde{P}_k^- - K_k H_k \tilde{P}_k^- \end{aligned} \tag{7.114}$$

These are the new Kalman gain equation and covariance-update equations. The state-update equations remain as before:

$$\begin{aligned} \hat{x}_k^- &= F_{k-1} \hat{x}_{k-1}^+ + G_{k-1} u_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - H_k \hat{x}_k^-) \end{aligned} \tag{7.115}$$

We see that the fading-memory filter is identical to the standard Kalman filter, with the exception that the time-update equation for the computation of the *a priori* estimation-error covariance has an α^2 factor in its first term. This serves to increase the uncertainty in the state estimate, which results in the filter giving more credence to the measurement. This is equivalent to increasing the process noise, which also results in the filter giving relatively more credence to the measurement. This strategy, along with other solutions to the filter divergence problem, was suggested early in the history of the Kalman filter [Sch67, Sor71a]. The fading-memory filter can be summarized as follows.

The fading-memory filter

1. The dynamic system is given by the following equations:

$$\begin{aligned} x_k &= F_{k-1}x_{k-1} + G_{k-1}u_{k-1} + w_{k-1} \\ y_k &= H_k x_k + v_k \\ E(w_k w_j^T) &= Q_k \delta_{k-j} \\ E(v_k v_j^T) &= R_k \delta_{k-j} \\ E(w_k v_j^T) &= 0 \end{aligned} \tag{7.116}$$

2. The Kalman filter is initialized as follows:

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ \tilde{P}_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T] \end{aligned} \tag{7.117}$$

3. Choose $\alpha \geq 1$ based on how much you want the filter to forget past measurements. If $\alpha = 1$ then the fading-memory filter is equivalent to the standard Kalman filter. In most applications, α is only slightly greater than 1 (for example, $\alpha \approx 1.01$).

4. The fading-memory filter is given by the following equations, which are computed for each time step $k = 1, 2, \cdots$:

$$\begin{aligned} \tilde{P}_k^- &= \alpha^2 F_{k-1}\tilde{P}_{k-1}^+ F_{k-1}^T + Q_{k-1} \\ K_k &= \tilde{P}_k^- H_k^T (H_k \tilde{P}_k^- H_k^T + R_k)^{-1} \\ &= \tilde{P}_k^+ H_k^T R_k^{-1} \\ \hat{x}_k^- &= F_{k-1}\hat{x}_{k-1}^+ + G_{k-1}u_{k-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k \hat{x}_k^-) \\ \tilde{P}_k^+ &= (I - K_k H_k)\tilde{P}_k^- (I - K_k H_k)^T + K_k R_k K_k^T \\ &= \left[(\tilde{P}_k^-)^{-1} + H_k^T R_k^{-1} H_k\right]^{-1} \\ &= \tilde{P}_k^- - K_k H_k \tilde{P}_k^- \end{aligned} \tag{7.118}$$

Note that $\tilde{P}$ is *not* equal to the covariance of the estimation error. However, the fading-memory filter is more robust to modeling errors than the standard Kalman filter.

■ **EXAMPLE 7.10**

In this example, we will show how the fading-memory filter makes the Kalman filter more responsive to measurements when the process noise is zero. Consider the following scalar system:

$$\begin{aligned} x_k &= x_{k-1} \\ y_k &= x_k + v_k \\ v_k &\sim (0, R) \end{aligned} \tag{7.119}$$

In other words, we are trying to estimate a constant on the basis of noisy measurements of that constant. Applying the fading-memory filter equations given in Equation (7.118) to this problem, we see that

$$\begin{aligned} P_k^- &= \alpha^2 P_{k-1}^+ \\ K_k &= \frac{P_k^-}{P_k^- + R} \\ &= \frac{\alpha^2 P_{k-1}^+}{\alpha^2 P_{k-1}^+ + R} \\ P_k^+ &= P_k^- - K_k H_k P_k^- \\ &= \alpha^2 P_{k-1}^+ - \left(\frac{\alpha^2 P_{k-1}^+}{\alpha^2 P_{k-1}^+ + R} \right) \alpha^2 P_{k-1}^+ \end{aligned} \tag{7.120}$$

As the filter approaches steady state, P_k^+ approaches a steady-state value that can be obtained from the above equation as

$$P_\infty^+ = \alpha^2 P_\infty^+ - \left(\frac{\alpha^2 P_\infty^+}{\alpha^2 P_\infty^+ + R} \right) \alpha^2 P_\infty^+ \tag{7.121}$$

This can be solved for P_∞^+ as

$$P_\infty^+ = \frac{(\alpha^2 - 1)R}{\alpha^2} \tag{7.122}$$

The steady-state Kalman gain K_∞ can then be solved as

$$\begin{aligned} K_\infty &= \frac{\alpha^2 P_\infty^+}{\alpha^2 P_\infty^+ + R} \\ &= \frac{\alpha^2 - 1}{\alpha^2} \end{aligned} \tag{7.123}$$

We see that if $\alpha = 1$ (i.e., if we use the standard Kalman filter) then $P_\infty^+ = K_\infty = 0$. However, if $\alpha > 1$ (i.e., if we use the fading-memory Kalman filter) then P_∞^+ and K_∞ will both be greater than zero. The measurement update equation for the state is given as

$$\hat{x}_k^+ = \hat{x}_k^- + K_k(y_k - \hat{x}_k^-) \tag{7.124}$$

For the standard Kalman filter, $\lim_{k\to\infty} K_k = 0$, which means that new measurements will be ignored and will not be used to update the state estimate.